

Overview in General

This file contains a general overview of the data in the graph including node labels and relationships types.

References

- [jqassistant](#)
- [Neo4j Python Driver](#)

Node Labels

Table 1a - Highest node count by label combination

Lists the 30 label combinations with the highest number of nodes. The labels with the lowest node count are listed in table 1b. The total list would sum up to the total number of labels (100%).

The whole table can be found in the CSV report `Node_label_combination_count`.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Git, Change, Update]	138781	46.077865
1	[Git, Change, Create]	44929	14.917261
2	[Git, Change, Delete]	22781	7.563714
3	[Java, ByteCode, Method, Member]	13492	4.479594
4	[Java, ByteCode, Parameter]	13306	4.417839
5	[Git, Commit]	11390	3.781691
6	[File, Git]	8286	2.751106
7	[Java, ByteCode, ParameterizedType, Bound]	7286	2.419087
8	[Java, ByteCode, Bound]	7245	2.405474
9	[Git, Change, Rename]	7018	2.330106
10	[Java, ByteCode, Member, Field]	3606	1.197259
11	[Java, ByteCode, Bound, WildcardType]	2950	0.979455
12	[Java, Value, ByteCode, Annotation]	2936	0.974806
13	[Xml, Element]	2162	0.717824
14	[Java, ByteCode, Constructor, Method, Member]	2114	0.701887
15	[Xml, Text]	1450	0.481427
16	[Java, ByteCode, Bound, TypeVariable]	1111	0.368873
17	[Java, ByteCode, Method, Member, Lambda]	968	0.321394
18	[Type, File, Java, ByteCode, ResolvedDuplicate...	890	0.295497
19	[Type, File, Java, Class, ByteCode]	845	0.280556
20	[Json, Key]	739	0.245362
21	[Value, Json, Scalar]	723	0.240049
22	[Java, Value, ByteCode, Primitive]	673	0.223448
23	[Type, File, Java, ByteCode, JavaType]	649	0.215480
24	[Java, ByteCode, Method, Member, GenericDeclar...	578	0.191907
25	[Type, File, Java, ByteCode, ExternalType]	412	0.136792
26	[Git, Change, Copy]	298	0.098942
27	[Author, Git, Person]	295	0.097945
28	[Value, Array]	289	0.095953
29	[Committer, Git, Person]	242	0.080348

Chart 1a - Highest node count by label combination

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in Chart 1b.

<Figure size 640x480 with 0 Axes>

Nodes per label combination (more than 0.5% overall)

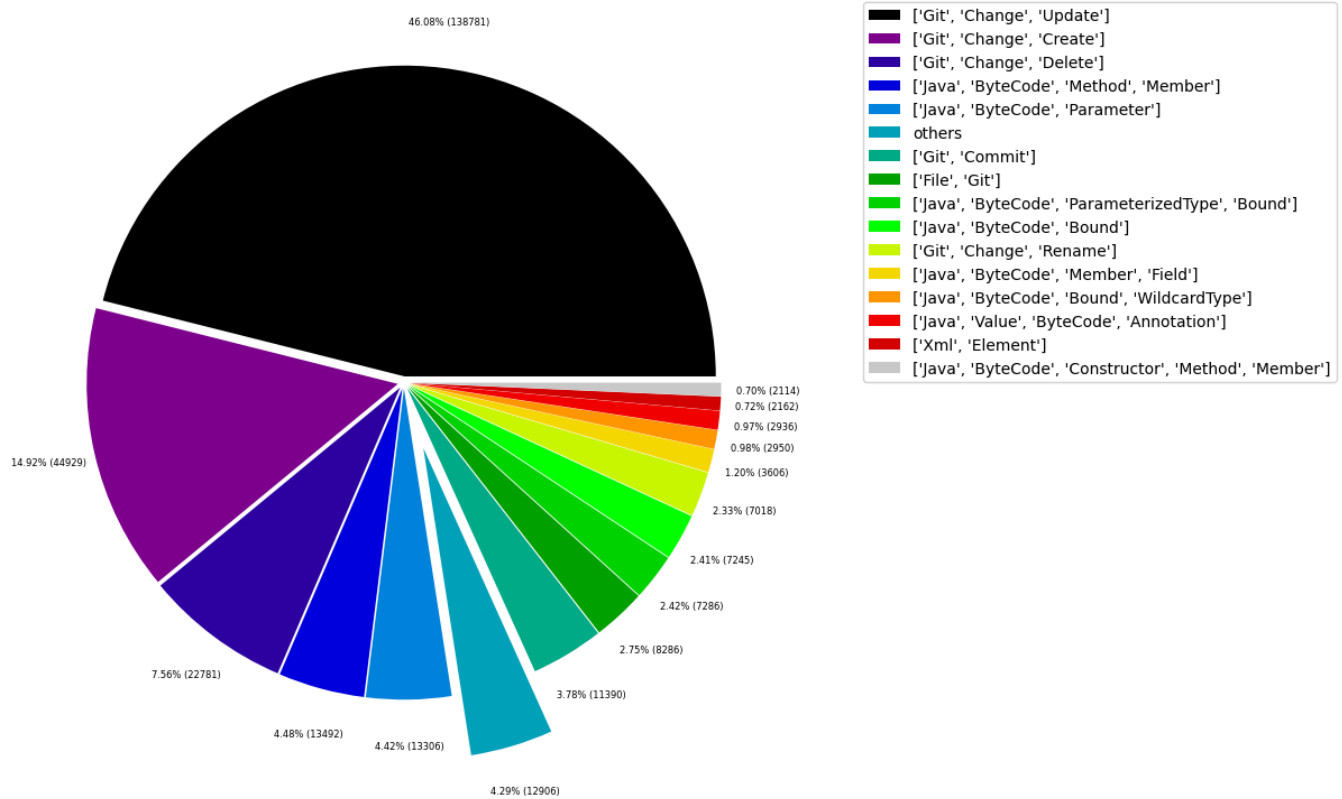


Table 1b - Lowest node count by label combination

Lists the 30 label combinations with the lowest number of nodes until they reach 0.5% of the total node count, which are shown above.

	nodeLabels	nodesWithThatLabels	nodesWithThatLabelsPercent
0	[Analyze, Task, jQAssistant]	1	0.000332
1	[Git, Branch]	1	0.000332
2	[Repository, File, Git]	1	0.000332
3	[File, Json]	2	0.000664
4	[File]	3	0.000996
5	[Java, ByteCode, Constructor, Method, Member, ...]	4	0.001328
6	[Maven, Exclusion]	5	0.001660
7	[Value, Array, Json]	6	0.001992
8	[File, Maven, Xml, Pom, Document]	9	0.002988
9	[Type, File, Java, ByteCode, Void]	9	0.002988
10	[Java, ManifestSection]	9	0.002988
11	[File, Artifact, Jar, Archive, Zip, Java]	9	0.002988
12	[File, Java, Manifest]	9	0.002988
13	[File, Java, ServiceLoader]	10	0.003320
14	[File, Java, Properties]	12	0.003984
15	[Maven, ExecutionGoal]	16	0.005312
16	[Maven, PluginExecution]	16	0.005312
17	[Xml, Attribute]	18	0.005976
18	[jQAssistant, Rule, Concept]	19	0.006308
19	[Type, File, Java, ByteCode, Throwable, Extern...	21	0.006972
20	[Maven, Plugin]	21	0.006972
21	[Maven, Configuration]	21	0.006972
22	[Type, File, Java, ByteCode, Throwable, Resolv...	22	0.007304
23	[Type, File, Java, ByteCode, Enum]	28	0.009297
24	[Type, File, Java, ByteCode, PrimitiveType]	30	0.009961
25	[Xml, Namespace]	36	0.011953
26	[Type, File, Java, ByteCode, Annotation]	44	0.014609
27	[File, Directory]	44	0.014609
28	[Type, File, Java, Class, ByteCode, Throwable]	55	0.018261
29	[Type, File, Java, ByteCode, Interface, Generi...	85	0.028222

Chart 1b - Lowest node count by label combination

Shows the lowest (less than 0.5% overall) node count label combinations. Therefore, this plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

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Nodes per label combination (less than 0.5% overall)

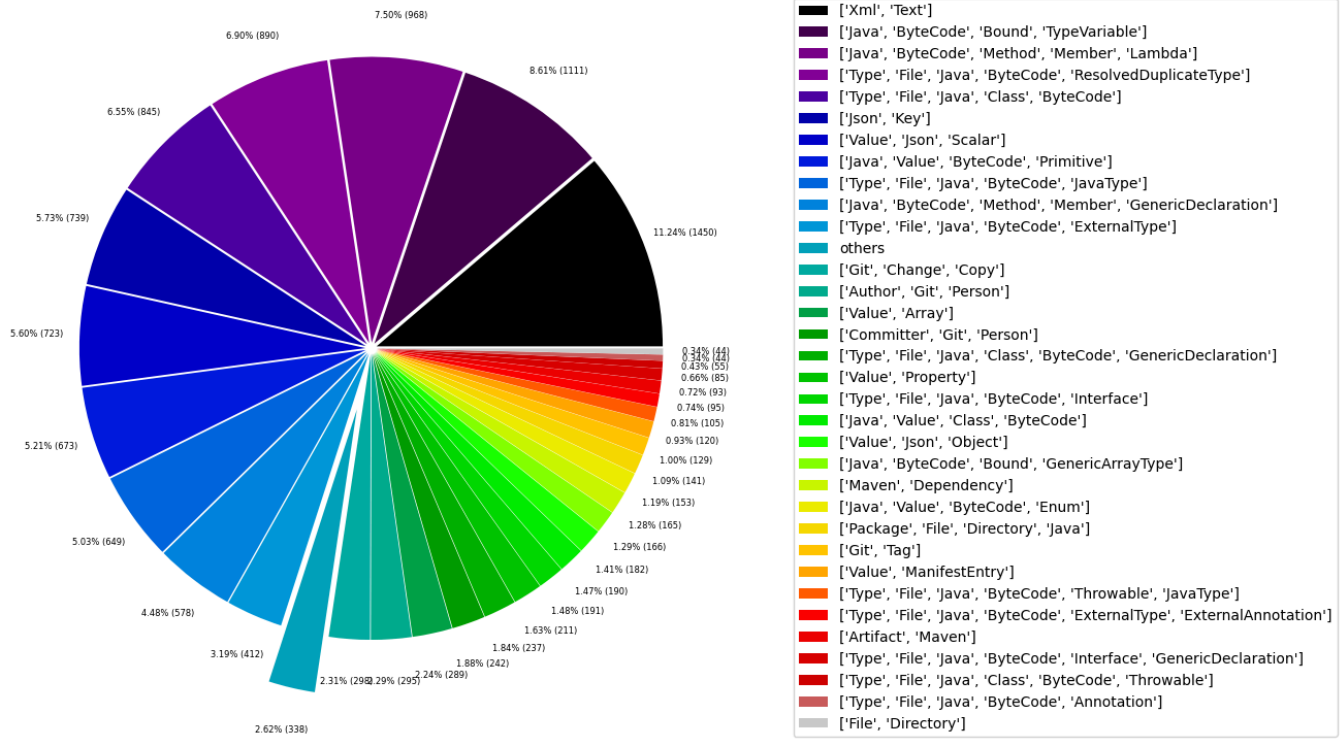


Table 1c - Highest node count by single label

Lists the 40 labels with the highest number of nodes. Doesn't sum up to the total number of nodes or 100% because one node can have multiple labels. Helps to identify commonly used labels.

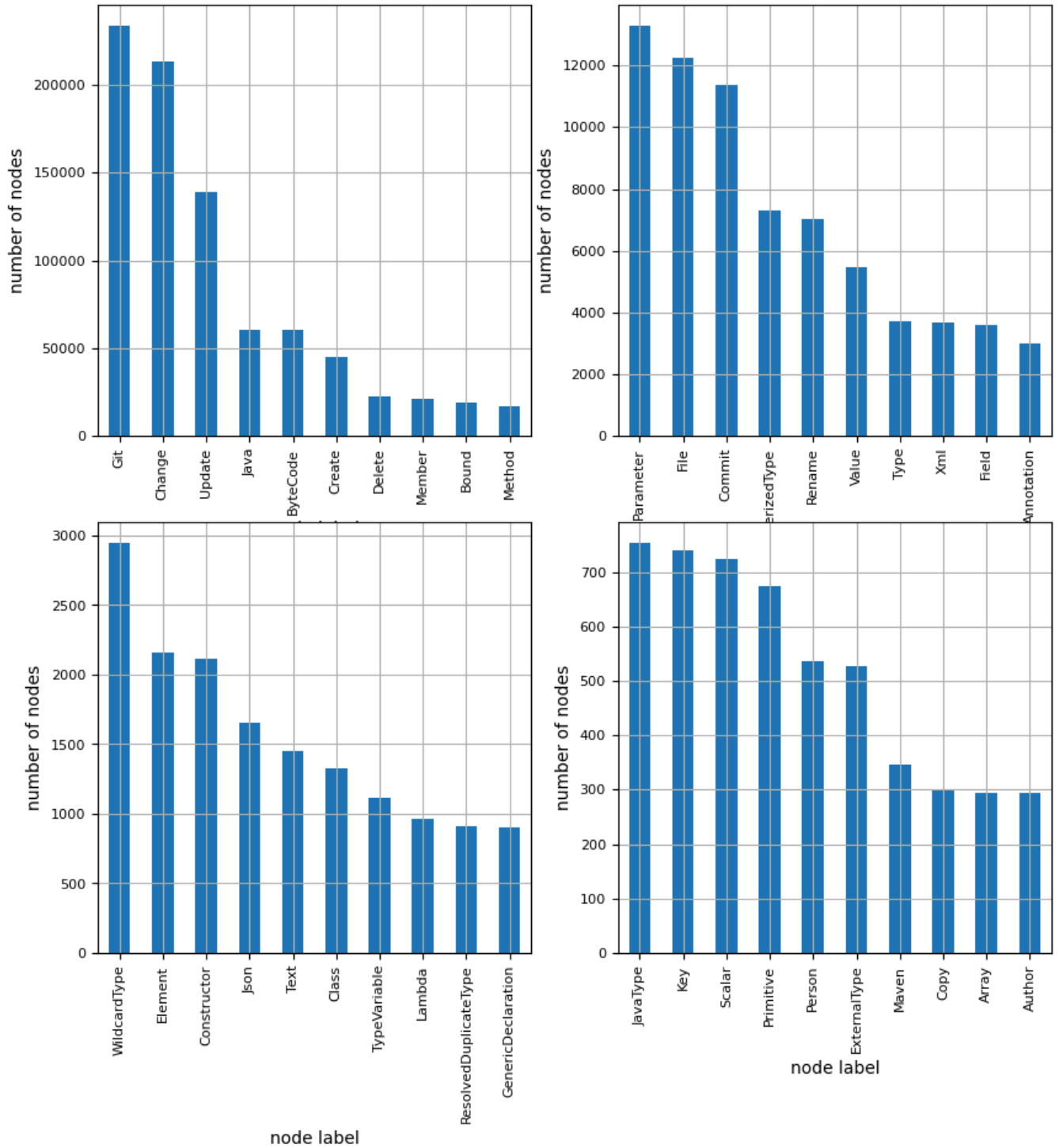
	nodeLabel	nodesWithThatLabel	nodesWithThatLabelPercent
0	Git	234151	77.742473
1	Change	213807	70.987888
2	Update	138781	46.077865
3	Java	60686	20.148877
4	ByteCode	60496	20.085794
5	Create	44929	14.917261
6	Delete	22781	7.563714
7	Member	20762	6.893369
8	Bound	18758	6.228004
9	Method	17156	5.696110
10	Parameter	13306	4.417839
11	File	12244	4.065235
12	Commit	11390	3.781691
13	ParameterizedType	7286	2.419087
14	Rename	7018	2.330106
15	Value	5483	1.820458
16	Type	3718	1.234445
17	Xml	3675	1.220168
18	Field	3606	1.197259
19	Annotation	2980	0.989415
20	WildcardType	2950	0.979455
21	Element	2162	0.717824
22	Constructor	2118	0.703215
23	Json	1652	0.548495
24	Text	1450	0.481427
25	Class	1327	0.440589
26	TypeVariable	1111	0.368873
27	Lambda	968	0.321394
28	ResolvedDuplicateType	912	0.302801
29	GenericDeclaration	904	0.300145
30	JavaType	754	0.250342
31	Key	739	0.245362
32	Scalar	723	0.240049
33	Primitive	673	0.223448
34	Person	537	0.178294
35	ExternalType	528	0.175306
36	Maven	346	0.114878
37	Copy	298	0.098942
38	Array	295	0.097945
39	Author	295	0.097945

Chart 1c - Highest node count by label

Shows the 40 labels with the highest number of nodes.

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Node count by label



Relationship Types

Table 2a - Highest relationship count by type

Lists the 30 relationship types with the highest number of occurrences. The whole table can be found in the CSV report `Relationship_type_count`.

Total number of relationships: 936107

	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	CONTAINS_CHANGE	213807	22.840017
1	MODIFIES	213807	22.840017
2	UPDATES	138781	14.825335
3	CREATES	52245	5.581093
4	INVOKES	36801	3.931281
5	DELETES	29799	3.183290
6	COMMITTED	22780	2.433482
7	DEPENDS_ON	22504	2.403999
8	OF_TYPE	21945	2.344283
9	DECLARES	21222	2.267049
10	OF_RAW_TYPE	17367	1.855237
11	HAS	14444	1.542986
12	HAS_PARENT	13921	1.487116
13	RETURNS	12858	1.373561
14	HAS_COMMIT	11390	1.216741
15	READS	9382	1.002236
16	HAS_ACTUAL_TYPE_ARGUMENT	8414	0.898829
17	HAS_FILE	8286	0.885155
18	RENAMES	7018	0.749701
19	OF_GENERIC_TYPE	6000	0.640952
20	RESOLVES_TO	4236	0.452512
21	SIMILAR	4078	0.435634
22	WRITES	3948	0.421747
23	CONTAINS	3928	0.419610
24	HAS_NEW_NAME	3662	0.391195
25	RETURNS_GENERIC	3597	0.384251
26	CHANGED_TOGETHER_WITH	3034	0.324108
27	ANNOTATED_BY	2924	0.312357
28	REQUIRES	2233	0.238541
29	HAS_FIRST_CHILD	2162	0.230957

Chart 2a - Highest relationship count by type

Values under 0.5% will be grouped into "others" to get a cleaner plot. The group "others" is then broken down in the second chart.

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Relationship types (more than 0.5% overall)

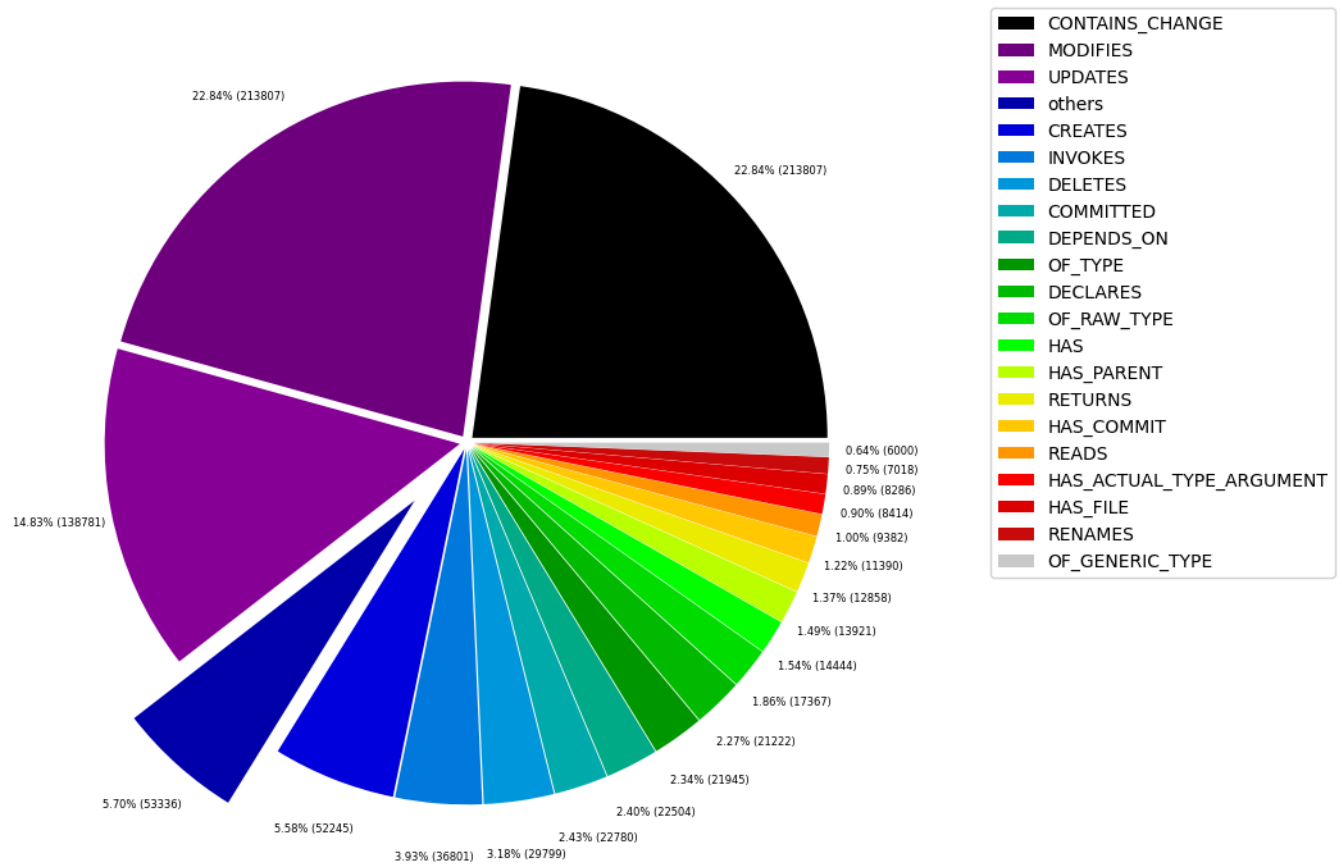


Table 2b - Lowest relationship count by type

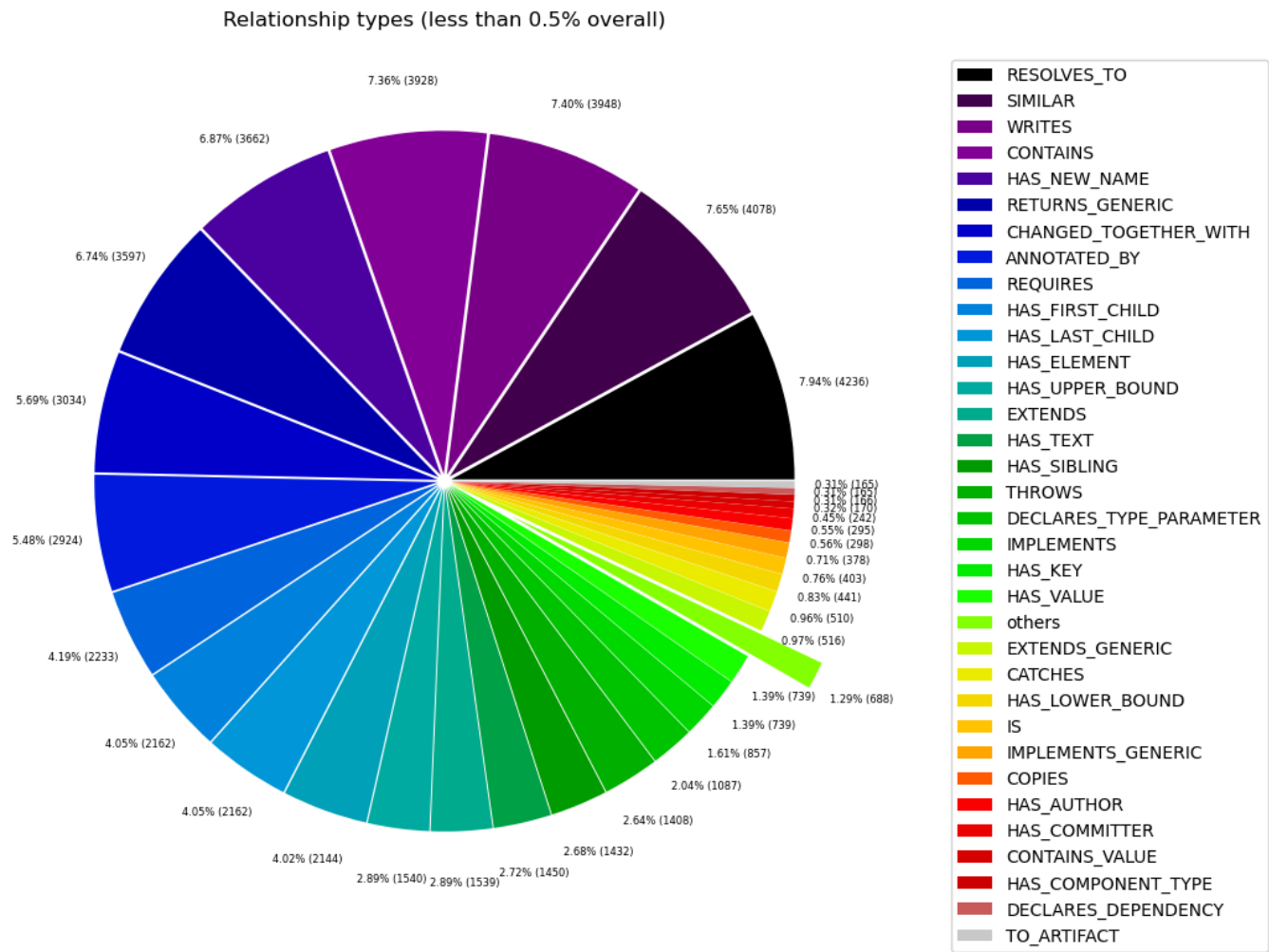
Lists the 30 relationships type with the lowest number of occurrences up to 0.5% of the total node count. This is essentially breaking down the "others" slice from the chart above.

	relationshipType	nodesWithThatRelationshipType	nodesWithThatRelationshipTypePercent
0	HAS_PROPERTY	1	0.000107
1	HAS_BRANCH	1	0.000107
2	HAS_HEAD	2	0.000214
3	THROWS_GENERIC	5	0.000534
4	EXCLUDES	5	0.000534
5	DESCRIBES	9	0.000961
6	HAS_ROOT_ELEMENT	11	0.001175
7	HAS_GOAL	16	0.001709
8	HAS_EXECUTION	16	0.001709
9	OF_NAMESPACE	18	0.001923
10	HAS_ATTRIBUTE	18	0.001923
11	INCLUDES_CONCEPT	19	0.002030
12	USES_PLUGIN	21	0.002243
13	IS_ARTIFACT	21	0.002243
14	HAS_CONFIGURATION	21	0.002243
15	REQUIRES_TYPE_PARAMETER	24	0.002564
16	REQUIRES_CONCEPT	28	0.002991
17	DECLARES_NAMESPACE	36	0.003846
18	HAS_DEFAULT	39	0.004166
19	COPY_OF	119	0.012712
20	HAS_TAG	129	0.013780
21	ON_COMMIT	129	0.013780
22	DECLARES_DEPENDENCY	165	0.017626
23	TO_ARTIFACT	165	0.017626
24	HAS_COMPONENT_TYPE	166	0.017733
25	CONTAINS_VALUE	170	0.018160
26	HAS_COMMITTER	242	0.025852
27	HAS_AUTHOR	295	0.031513
28	COPIES	298	0.031834
29	IMPLEMENTS_GENERIC	378	0.040380

Chart 2b - Lowest relationship count by type

Shows the lowest (less than 0.5% overall) relationship types. This plot breaks down the "others" slice of the pie chart above. Values under 0.01% will be grouped into "others" to get a cleaner plot.

<Figure size 640x480 with 0 Axes>



Node labels with their relationships

Table 3a - Highest relationship count by node labels and relationship type

Lists the 30 node labels and their relationship types with the highest number of occurrences.

	sourceLabels	relationType	targetLabels	numberOfRelationships	numberOfNodesWithSameLabelsAsSource	numbe
0	[Git, Change, Update]	UPDATES	[File, Git]	138781	138781	
1	[Git, Change, Update]	MODIFIES	[File, Git]	138781	138781	
2	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Update]	138781	11390	
3	[Git, Change, Create]	MODIFIES	[File, Git]	44929	44929	
4	[Git, Change, Create]	CREATES	[File, Git]	44929	44929	
5	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Create]	44929	11390	
6	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Delete]	22781	11390	
7	[Git, Change, Delete]	DELETES	[File, Git]	22781	22781	
8	[Git, Change, Delete]	MODIFIES	[File, Git]	22781	22781	
9	[Java, ByteCode, Method, Member]	INVOKES	[Java, ByteCode, Method, Member]	22320	13394	
10	[Git, Commit]	HAS_PARENT	[Git, Commit]	13912	11390	
11	[Repository, File, Git]	HAS_COMMIT	[Git, Commit]	11390	1	
12	[Committer, Git, Person]	COMMITTED	[Git, Commit]	11390	242	
13	[Author, Git, Person]	COMMITTED	[Git, Commit]	11390	295	
14	[Java, ByteCode, Method, Member]	HAS	[Java, ByteCode, Parameter]	8489	13394	
15	[Java, ByteCode, Method, Member]	READS	[Java, ByteCode, Member, Field]	8453	13394	
16	[Repository, File, Git]	HAS_FILE	[File, Git]	8286	1	
17	[Git, Commit]	CONTAINS_CHANGE	[Git, Change, Rename]	7018	11390	
18	[Git, Change, Rename]	CREATES	[File, Git]	7018	7018	
19	[Git, Change, Rename]	RENAMES	[File, Git]	7018	7018	
20	[Git, Change, Rename]	DELETES	[File, Git]	7018	7018	
21	[Git, Change, Rename]	MODIFIES	[File, Git]	7018	7018	
22	[Java, ByteCode, Parameter]	OF_TYPE	[Type, File, Java, ByteCode, JavaType]	6160	13306	
23	[File, Git]	HAS_NEW_NAME	[File, Git]	3662	8286	
24	[Java, ByteCode, Bound]	OF_RAW_TYPE	[Type, File, Java, ByteCode, JavaType]	3466	7245	
25	[Java, ByteCode, ParameterizedType, Bound]	OF_RAW_TYPE	[Type, File, Java, ByteCode, JavaType]	3114	7286	
26	[Java, ByteCode, ParameterizedType, Bound]	HAS_ACTUAL_TYPE_ARGUMENT	[Java, ByteCode, Bound, WildcardType]	2950	7286	
27	[Java, ByteCode, Parameter]	OF_GENERIC_TYPE	[Java, ByteCode, ParameterizedType, Bound]	2698	13306	
28	[Java, ByteCode, Constructor, Method, Member]	WRITES	[Java, ByteCode, Member, Field]	2520	2114	
29	[Java, ByteCode, ParameterizedType, Bound]	HAS_ACTUAL_TYPE_ARGUMENT	[Java, ByteCode, Bound, TypeVariable]	2475	7286	

Graph Density

```
total_number_of_nodes (vertices): 301188
total_number_of_relationships (edges): 936107
-> total directed graph density: 1.0319332485946026e-05
-> total directed graph density in percent: 0.0010319332485946027
```